

Nicolás Aller Ponte

A Coruña, Spain | nicooaller04@gmail.com | 604 00 65 36 | nicolasallerponte.com
linkedin.com/in/nicolasallerponte | github.com/nicolasallerponte

Education

Universidade da Coruña, in Bachelor's Degree in Data Science and Engineering – A Coruña, Spain Sept 2022 – present

- Knowledge of the complete data lifecycle: HPC, statistics, data engineering, databases and AI.

Experience

Research Assistant, Inditex-UDC Chair of AI in Green Algorithms – A Coruña, Spain Nov 2025 – Mar 2026

- Research and development of efficient algorithms for AI models.

AI Engineer, Gradiant – A Coruña, Spain Sept 2025 – Oct 2025

- Development of a car damage detection and parts evaluation system.

Languages

Proficiency: Spanish (Native), Galician (Native), English (Intermediate), German (Basic)

Skills

Data Engineering & Cloud: Python, R, SQL, MongoDB, Docker, Git, Spark, AWS, GCP

AI & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Computer Vision, NLP

HPC & Systems Programming: C, C++, Parallel Computing (OpenMP, MPI), Computer Architecture, CUDA

Awards

First Place - Best Use of Open Source AI (HackUDC 2026): Recognition for the integration and optimization of open-source models (SigLIP, DINOv2) in a scalable production environment

Second Place - Inditex-Tech Challenge (HackUDC 2026): Finalist for developing a visual search engine for massive fashion catalogs

Second Place - GDG Challenge (Impacthon 2026): Runner-up for developing LocalFold, a web portal for protein structure prediction using AlphaFold2, A2A multi-agent system, and Gemini-powered analysis on Google Cloud

Projects

Outfit Intelligence - HackUDC – A Coruña, Spain Mar 2026

- Multi-layer visual search pipeline using YOLOv8 and SigLIP-384 to identify catalog products in outfit images.
- Model fine-tuning with contrastive loss and cross-validation, combining SigLIP and DINOv2 in an ensemble to improve accuracy.
- Recommendation engine using FAISS and a REST API with FastAPI on a catalog of 27,000+ products.

LocalFold - Impacthon 2026 – GDG Challenge Apr 2026

- Web portal for protein structure prediction with AlphaFold2 without HPC knowledge, with interactive 3D visualization (Mol*).
- A2A multi-agent system with four specialized agents using Gemini with Google Search Grounding.
- Deployed on Google Cloud Run with FastAPI, Firestore, and multilingual support.